

Custom Furniture Workshop Database Project

1. Project Overview

For my individual database project, I chose a custom furniture workshop as my mini world. I picked this type of business because a furniture workshop has several things that need to be organized at the same time, such as customers, custom orders, employees, furniture items, materials, and payments.

The purpose of the database is to keep all of this information organized in one place. Instead of keeping customer information, order details, material information, and payments in separate notes or files, the database connects the information together. This makes it easier for the workshop to keep track of an order from the time a customer places it until the furniture is completed and paid for.

2. How I Determined the Requirements

I started by thinking about how a small custom furniture business would normally operate. A customer first contacts the workshop and places an order. An employee is responsible for handling the order. The order can include one or more furniture items, such as a table, desk, chair, or cabinet. Materials are needed to build the order, and the customer can make one or more payments.

From that process, I identified the main information that the database would need to store. I then separated the information into tables so that the same information would not have to be repeated unnecessarily. I also thought about the questions a workshop owner might want the database to answer, such as which orders are still in progress, which materials are running low, and how much a customer still owes.

3. Main Database Requirements

The database needs to store customer contact information, employee information, custom order details, furniture item details, available materials, materials needed for each order, and customer payments. Each important record needs its own ID so that it can be identified correctly.

The database also needs relationships between the tables. For example, an order must be connected to a customer and an employee. Furniture items and payments must be connected to the correct order. Materials must also be connected to the orders that use them.

4. Business Rules

The business rules I used for the database are:

- One customer can place many custom orders, but each custom order belongs to one customer.
- One employee can handle many custom orders, but each custom order is assigned to one employee.
- One custom order can contain many furniture items, but each furniture item belongs to one order.
- One custom order can have multiple payments, but each payment belongs to one order.
- One custom order can use many materials, and the same material can be used for many different orders.
- The amount of a material needed for a specific order is stored as `quantity_needed`.
- Customer email addresses are unique when an email address is provided.

- New orders have a default status of Pending.
- Each table uses a primary key to identify its records.

5. Tables and Their Purpose

Customer: Stores the customer's name, phone number, email address, and address.

Employee: Stores information about employees who work on or manage customer orders.

Custom_Order: Stores the main information about an order, including the customer, assigned employee, order date, expected completion date, status, total price, and notes.

Furniture_Item: Stores the individual furniture pieces included in an order. It includes information such as the item type, dimensions, wood type, finish, quantity, and price.

Material: Stores materials used by the workshop, including the material name, type, unit, cost, and quantity currently in stock.

Order_Material: Connects orders and materials. This table is needed because one order can use many materials and one material can be used for many orders. It also records the quantity needed.

Payment: Stores payments made toward an order, including the payment date, amount, and payment method.

6. Relationships

The Customer and Custom_Order tables have a one-to-many relationship because one customer can have multiple orders. Employee and Custom_Order also have a one-to-many relationship because an employee can handle multiple orders.

Custom_Order has a one-to-many relationship with Furniture_Item and Payment. The relationship between Custom_Order and Material is many-to-many, so I used the Order_Material junction table to connect them. This design keeps the data organized and avoids storing the same material information repeatedly for every order.

7. ER Diagram Design

I created two versions of the ER design. The first is a Chen-style ER diagram that shows the main entities, their attributes, relationships, and cardinalities. The Chen diagram will be drawn by hand.

I also created an EER/UML-style diagram in MySQL Workbench. The Workbench diagram shows the actual tables, primary keys, foreign keys, and relationships used in the database. The model contains seven tables: Customer, Employee, Custom_Order, Furniture_Item, Material, Order_Material, and Payment.

8. Database Creation in MySQL Workbench

After planning the tables and relationships, I created the database in MySQL Workbench. The database is named custom_furniture_workshop_db. I wrote a SQL script that creates the database and all seven tables.

Primary keys are used to uniquely identify records. Foreign keys connect related tables. I also used constraints such as NOT NULL, UNIQUE, and DEFAULT where they made sense. After creating the tables, I added sample data so I could test the database and make sure the relationships and queries worked correctly.

9. Testing the Database

I tested the database by running SELECT statements and other queries in MySQL Workbench. I checked the individual tables to make sure the sample records were inserted correctly.

I also tested queries that combine information from multiple tables. Examples include showing an order with its customer and assigned employee, showing materials needed for an order, calculating how much has been paid toward an order, finding orders that are still in progress, and finding materials with low stock. The queries returned results successfully, which showed that the database and relationships were working as expected.

10. Design Decisions and Challenges

One important design decision was creating Order_Material as a separate table. At first, materials might seem like they could be stored directly in the order table. However, one order can require several materials, and the same material can be used for many orders. The junction table solves this problem and also lets the database store the quantity needed for each material.

Another decision was separating Furniture_Item from Custom_Order. This allows one order to contain multiple pieces of furniture without repeating the customer's order information for every item.

While working on the project, I also had to make sure MySQL Workbench and the MySQL server were connecting correctly. After resolving the connection issue, I was able to run the database script, view all seven tables, test the queries, and generate the EER diagram.

11. What the Database Can Be Used For

This database could help a custom furniture workshop keep better records of its daily work. Employees could look up customer orders, check which furniture items are included, see what materials are required, review payment information, and check the status of an order.

The database could also help the business identify materials that need to be reordered and determine whether a customer still has a balance remaining on an order.

12. Conclusion

This project helped me understand how a real business process can be turned into a relational database. I learned that designing the tables first is important because the relationships determine how the information connects.

The final database uses seven related tables to organize customers, employees, orders, furniture items, materials, and payments. Using primary keys, foreign keys, and a junction table helped keep the design organized and reduced unnecessary repetition. After testing the database in MySQL Workbench, I was able to confirm that the tables and queries work together as intended.

